1 Scaling Law

1.1 Isobaric Core Radius

Goal: Similar whole mantle temperature for 2D and 3D.

R_{planet} = Radius of the planet

R_{core} = Radius of the core

r_{sphere} = Radius of isobaric core = $\sqrt[3]{3} \times r_{\text{imp}}$

r_{circle} =?

$$\frac{A_{\text{ic}}}{A_{\text{planet}} - A_{\text{core}}} = \frac{V_{\text{ic}}}{V_{\text{planet}} - V_{\text{core}}} \tag{1}$$

$$\frac{\pi r_{\text{circle}}^2}{\pi(R_{\text{planet}}^2 - R_{\text{core}}^2)} = \frac{\frac{4}{3}\pi r_{\text{sphere}}^3}{\frac{4}{3}\pi(R_{\text{planet}}^3 - R_{\text{core}}^3)} \tag{2}$$

$$r_{\text{circle}} = \sqrt{r_{\text{sphere}}^3 \left(\frac{R_{\text{planet}}^2 - R_{\text{core}}^2}{R_{\text{planet}}^3 - R_{\text{core}}^3}\right)} \tag{3}$$

$$r_{\text{circle}} = \sqrt{\left(\sqrt[3]{3} \times r_{\text{imp}}\right)^3 \left(\frac{R_{\text{planet}}^2 - R_{\text{core}}^2}{R_{\text{planet}}^3 - R_{\text{core}}^3}\right)} \tag{4}$$

$$r_{\text{circle}} = r_{\text{imp}}^{\frac{3}{2}} \times \sqrt{3 \left(\frac{R_{\text{planet}}^2 - R_{\text{core}}^2}{R_{\text{planet}}^3 - R_{\text{core}}^3}\right)} \tag{5}$$

$$r_{\text{circle}} = r_{\text{imp}}^{\frac{3}{2}} \times 0.00064967601 \tag{6}$$

1.2 Scale r_{imp}

$$r_{\text{circle}} = r_{\text{imp}}^{\frac{3}{2}} \times 0.00064967601 \tag{7}$$

$$\sqrt[3]{3} \times r'_{\text{imp}} = r_{\text{imp}}^{\frac{3}{2}} \times 0.00064967601 \tag{8}$$

$$r'_{\text{imp}} = r_{\text{imp}}^{\frac{3}{2}} \times 0.00045046019 \tag{9}$$